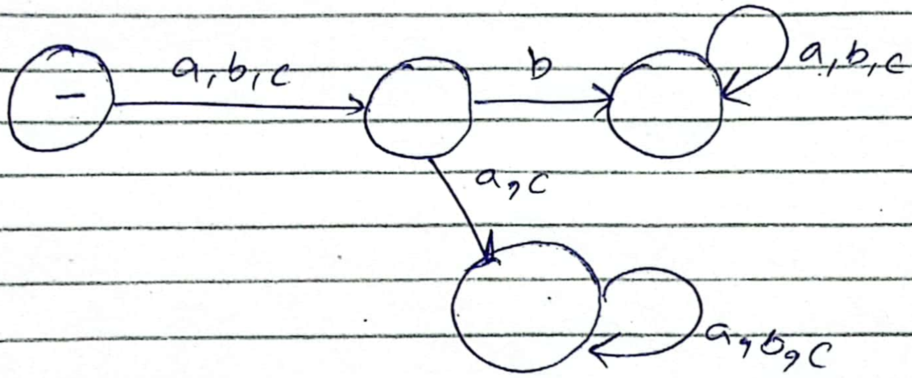
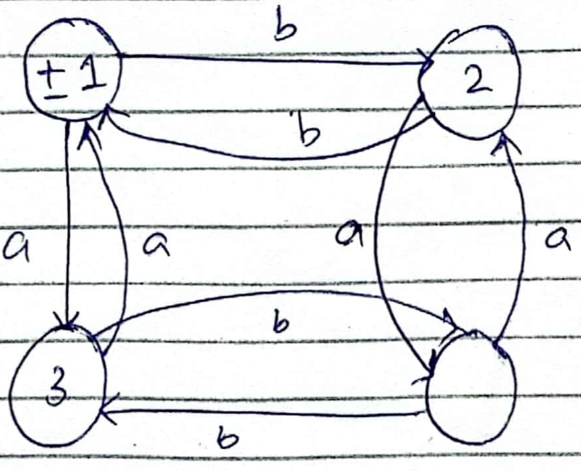
All words in which the second letter is b.

$\Sigma = \{a, b, c\}$

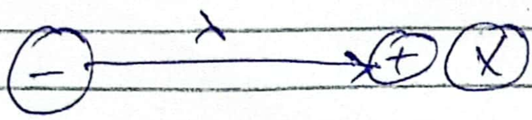
$L = \{ab, cb, bb, abc, bbc, cba \ldots\}$



All words having even number of a's and even number of b's



→ DFA cannot have a null transition.

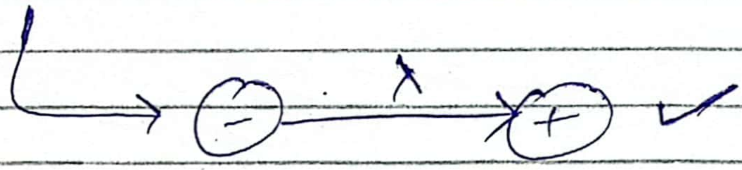


Thursday 31-10-2024

NFA

↳ Non deterministic finite Automata.

→ Exactly one start state.
→ One or more final states (May be none).
→ Single character while making transition.
→ Can have a ~~xx~~ duplicate edge.
→ Can have a missing edge.
→ Can have a null transition

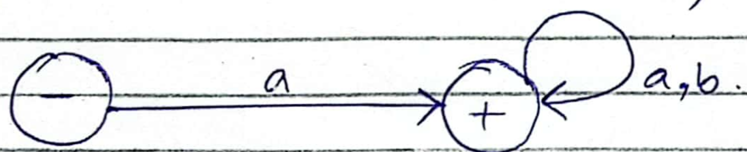


NFA:

All words that start with a.

$\Sigma = \{a, b\}$

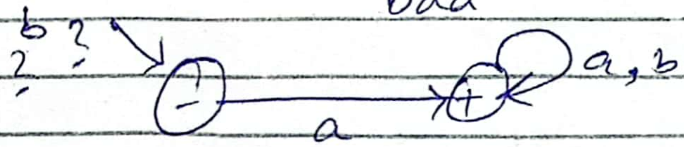
$L = \{a, ab, aab, aaaab - - - \}$



→ Ending at non final state → rejected.
→ if missing edge → reject

example.

baa

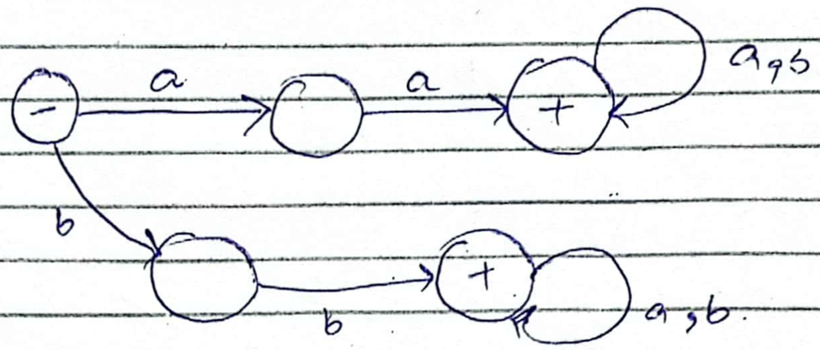


b not pressent so baa is rejected.

All words that start with double letter.
(NFA)

$\Sigma = \{a, b\}$

L = { aa, bb, aab, bba, aaaaba - - - }

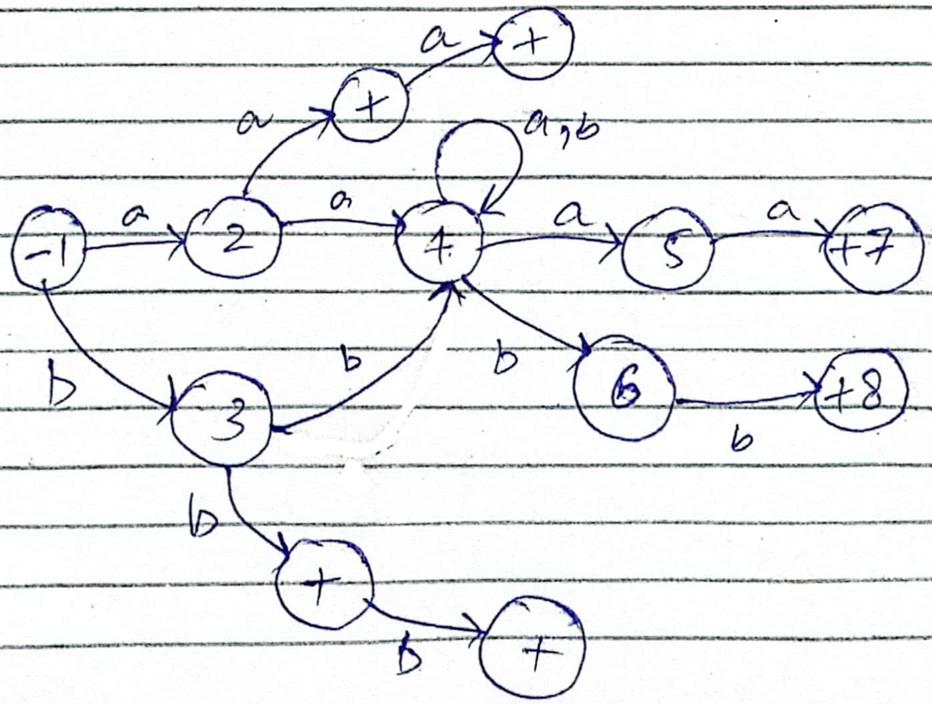


All words that start and end with double letter.

(NFA).

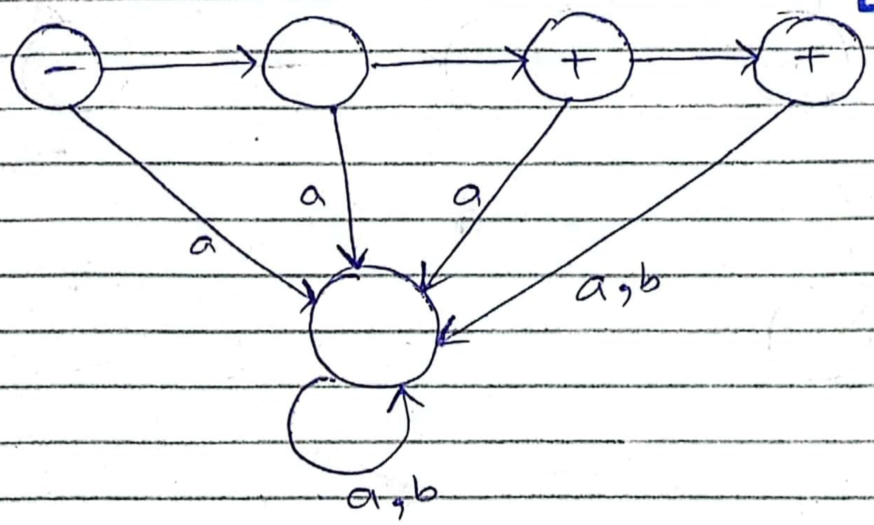
$\Sigma = \{a, b\}$

L = { aa, bb, aabb, bbaa - - -
aaa, bbb - - - }

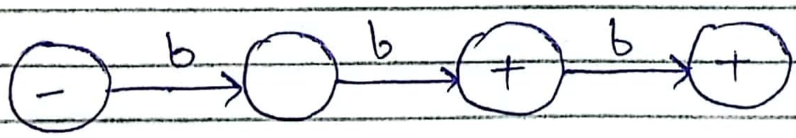


$\Sigma = \{ a, b \}$

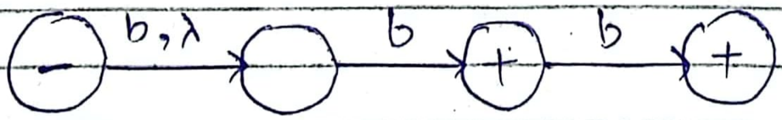
$L = \{ bb, bbb \}$ → DFA



NFA

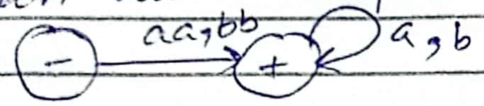


OR



Transition Graph.

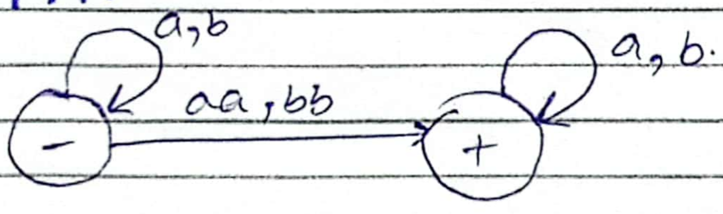
→ Can have a null transition.
→ Can have a missing edge.
→ can more than one start ~~state~~ state.
→ One or more final state (May be none)
→ string instead of a character can be read
→ can have duplicate edge.



→ All words that contain double letter.

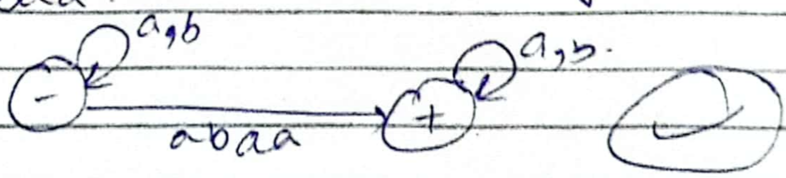
R.E = $(a+b)^*(aa+bb)(a+b)^*$

TFA:-



★ We can write whole string.

abaa.



if abaa is valid you can write it.